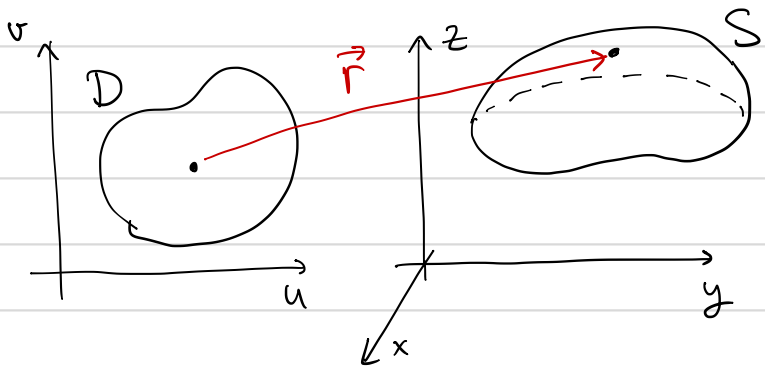
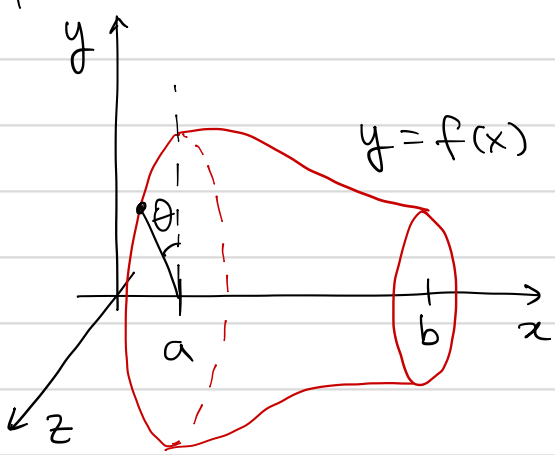
Parametrization of surfaces

(Book: 16.6)



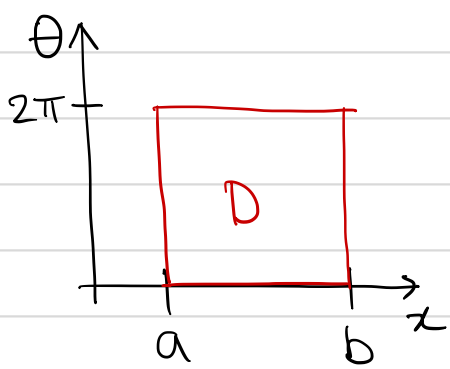
$$\vec{r}(u,v) = \langle x(u,v), y(u,v), z(u,v) \rangle$$

Example: Surfaces of revolution



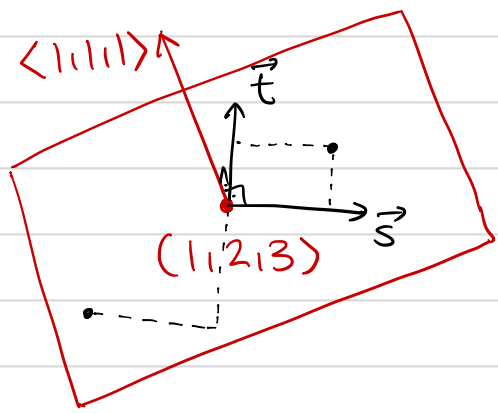
$$\vec{r}(x,\theta) = (x, f(x)\cos\theta, f(x)\sin\theta)$$

$$(a \le x \le b,\ 0 \le \theta \le 2\pi)$$



Problem: Parametrize the plane through $\langle 1,2,3 \rangle$ with normal vector $\langle 1,1,1 \rangle$.

Sol'n: Find two vectors perpendicular to $\langle 1,1,1 \rangle$:

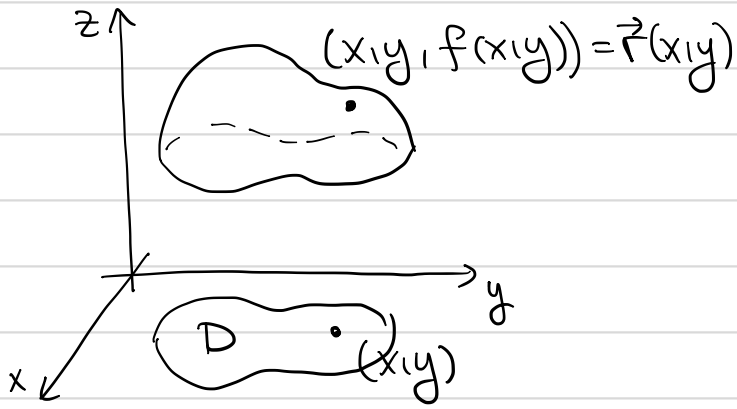


$$\vec{t} \cdot \langle 1,1,1 \rangle = 0 \qquad \vec{t} = \langle 1,-1,0 \rangle$$
$$\vec{s} \cdot \langle 1,1,1 \rangle = 0 \quad \Leftarrow \quad \vec{s} = \langle 1,0,-1 \rangle$$

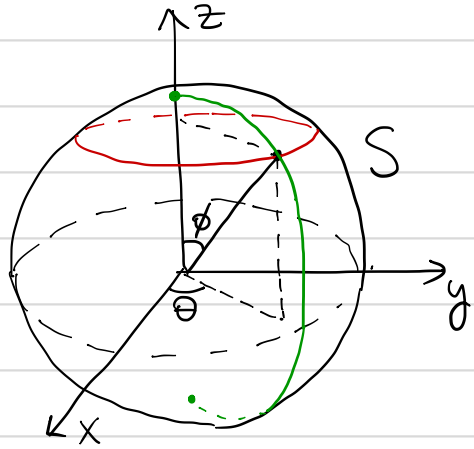
(for example)

$$\vec{r}(u,v) = \langle 1,2,3 \rangle + u\langle 1,-1,0 \rangle + v\langle 1,0,-1 \rangle$$
$$= \langle 1+u+v,\ 2-u,\ 3-v \rangle$$

Example: S = graph of $f(x,y)$ over D

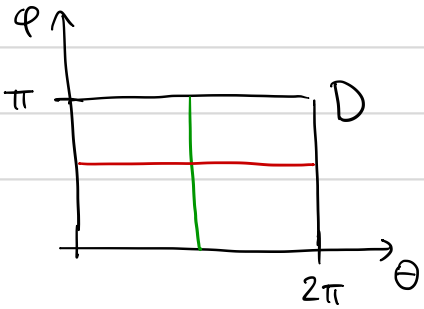


Example: S = Sphere of radius a

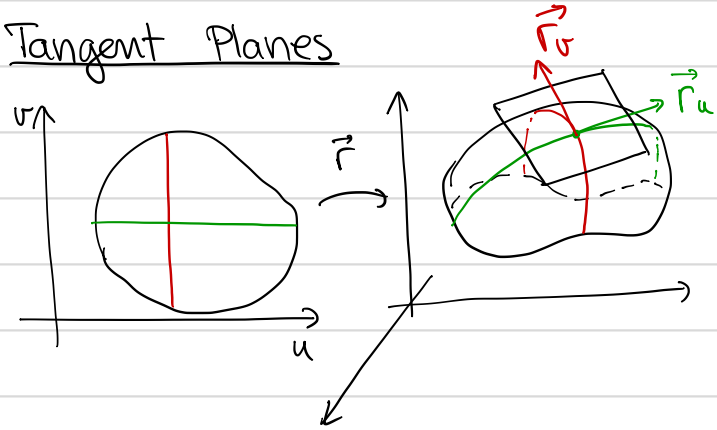


$$\vec{r}(\theta,\varphi) = \langle a \sin\varphi \cos\theta,\ a \sin\varphi \sin\theta,\ a\cos\varphi \rangle$$

$(0 \le \theta \le 2\pi,\ 0 \le \varphi \le \pi)$



Tangent Planes



$$\vec{r}_u \times \vec{r}_v = \frac{\partial \vec{r}}{\partial u}(u_0,v_0) \times \frac{\partial \vec{r}}{\partial v}(u_0,v_0)$$

is normal to the tangent plane of S at $\vec{r}(u_0,v_0)$.

Example:

S: $x = u^2 \quad y = v^2 \quad z = u + 2v$

Find: Tangent plane at $(1,1,3)$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2u & 0 & 1 \\ 0 & 2v & 2 \end{vmatrix}$$

$$= -2v\,\hat{i} - 4u\,\hat{j} + 4uv\,\hat{k}$$

Normal at $(1,1,3)$:

$\vec{N} = (-2,-4,4)$

Equation for tangent plane at $(1,1,3)$:

$-2(x-1) - 4(x-1) + 4(x-3) = 0$